

From shared evidence to a reliability-aware decision

The environment changes the evidence, watches the response, and reveals labels last

1. Decision setup

question → agents → consensus

Question: "Accept this multi-agent decision?"



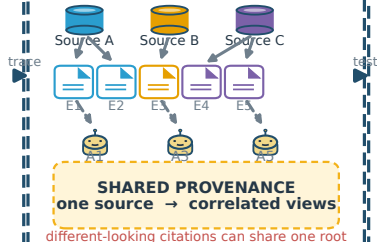
YES YES YES YES YES

HIGH CONSENSUS · 5/5 YES
consensus ≠ reliability



2. Provenance

source → evidence → agents



3. Counterfactual tests

controlled evidence intervention

Original: $E = \{E1, E2, E3\} \cdot A(E) = \text{YES}$

REMOVE **REVERSE** **SUBSTITUTE**

$E_j \rightarrow \emptyset$ $E_j \rightarrow \bar{E}_j$ $E_j \rightarrow E'_j$

YES → NO flip YES → YES inertia YES → NO flip

$$\Delta_{ij} = A_i(E) - A_i(E')$$

Does the decision respond to evidence?
fragility → false-consensus risk

observe

4. Reliability & routing

INTERVENTION MATRIX

A1	0	1	0
A2	0	1	1
A3	0	0	1
A4	0	1	0
A5	0	0	1

1 = evidence-sensitive 0 = suspicious inertia

$$\Delta_{ij} = A_i(E) - A_i(E')$$

green = flip · orange = inertia

INTERPRETABLE RISK ROUTER

answer-flip
inertia

complete
inertia

aggregate

$$R_{PI} / R_{\text{sym}}$$

pre-outcome risk

shared-source fraction · secondary signal

$$R_{PI} = 0.1D_{\text{inert}} + 0.3f_{\text{flip}} + 0.6f_{\text{shared}}$$

SELECTIVE ROUTING

LOW RISK
ACCEPT

HIGH RISK
ABSTAIN / ESCALATE

predict reliability before labels are revealed

BoolQ V12.1 · compact validation check

flip inertia: 0.807

shared source: 0.498

Risk@80 22.0% → 13.3%

selective routing reduces retained error at 80% coverage

HIGH CONSENSUS ≠ RELIABLE

INTERVENTION > CITATIONS